

Recent advances in kernel machines

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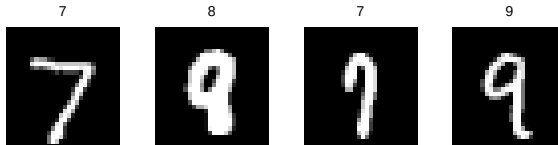


Optical character recognition

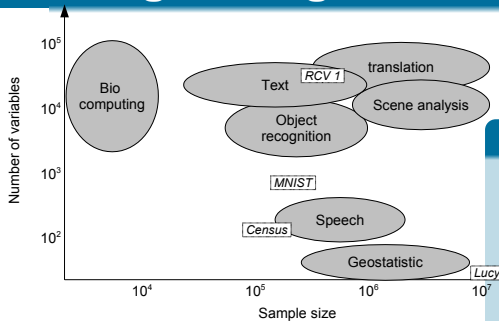
Example (The MNIST database)

- ▶ MNIST^a, data = « image-label »
- ▶ $n = 60,000$; $d = 700$; classes = 10
- ▶ Kernel error rate = 0.56 %,
- ▶ Best error rate = 0.4 % .

^a<http://yann.lecun.com/exdb/mnist/index.html>



Learning challenges: the size effect



3 key issues

1. learn any problem:
 - ▶ functional universality
2. from data:
 - ▶ statistical consistency
3. with large data sets:
 - ▶ computational efficiency

kernel machines address these three issues
(up to a certain point regarding efficiency)

Notations

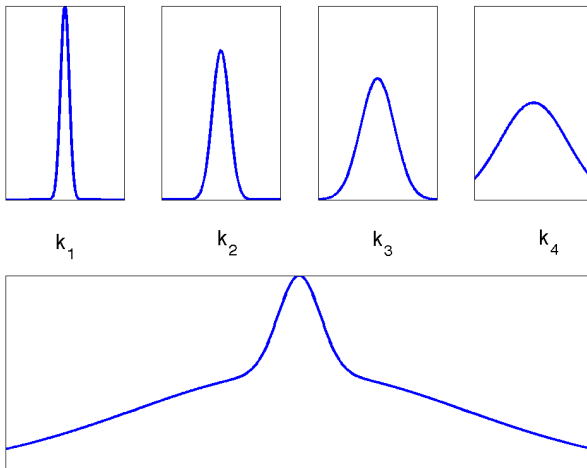
- ▶ inputs $\mathbf{x} \in \mathcal{X} = \mathbb{R}^d$, d features
- ▶ outputs y
- ▶ training set $(\mathbf{x}_i, y_i)$, $i = 1, n$, n exemples
- ▶ test set $(\mathbf{x}_j, y_j)$, $j = 1, \ell$, ℓ exemples
- ▶ kernel $k(\mathbf{x}_i, \mathbf{x}_j) \quad \mathcal{X} \times \mathcal{X} \mapsto \mathbb{R}$
- ▶ RKHS $\mathcal{H}$ (set of hypothesis associated with **positive** kernel k)
- ▶ RKKS $\mathcal{K}$ (set of hypothesis associated with kernel k) – Krein

Definition (Kernel machines)

$$\mathcal{A}((\mathbf{x}_i, y_i)_{i=1,n})(\mathbf{x}) = \psi\left(\sum_{i=1}^n \alpha_i k(\mathbf{x}, \mathbf{x}_i) + \sum_{j=1}^p \beta_j q_j(\mathbf{x})\right)$$

α et β : parameters to be estimated

Multiple Kernel: illustration



$$k = m_1 k_1 + m_2 k_2 + m_3 k_3 + m_4 k_4$$

$$m_2 = m_3 = 0$$

Multiple Kernel functional Learning

The problem (for given C and t)

$$\begin{aligned} \min_{\{f_m\}, b, \xi, d} \quad & \frac{1}{2} \sum_m \frac{1}{d_m} \|f_m\|_{\mathcal{H}_m}^2 + C \sum_i \xi_i \\ \text{s.t.} \quad & \left| \sum_m f_m(x_i) + b - y_i \right| \leq t + \xi_i \quad \forall i, \xi_i \geq 0 \quad \forall i \\ & \sum_m d_m = 1, \quad d_m \geq 0 \quad \forall m, \end{aligned}$$

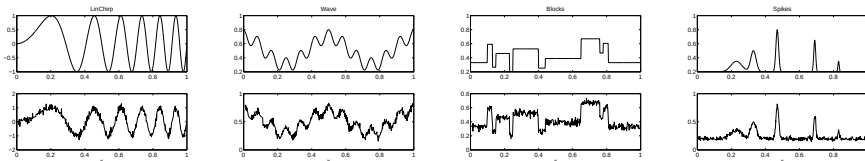
regularization formulation

$$\min_{\{f_m\}, b, d} \quad \frac{1}{2} \sum_m \frac{1}{d_m} \|f_m\|_{\mathcal{H}_m}^2 + C \sum_i \max(\left| \sum_m f_m(x_i) + b - y_i \right| - t, 0) \\ \sum_m d_m = 1, \quad d_m \geq 0 \quad \forall m,$$

Equivalently

$$\min_{\{f_m\}, b, \xi, d} \sum_i \max \left(\left| \sum_m f_m(x_i) + b - y_i \right| - t, 0 \right) + \frac{1}{2C} \sum_m \frac{1}{d_m} \|f_m\|_{\mathcal{H}_m}^2 + \mu \sum_m |d_m|$$

Multiple Kernel experiments



	Single Kernel	Kernel <i>Dil</i>		Kernel <i>Dil-Trans</i>	
Data Set	Norm. MSE (%)	#Kernel	Norm. MSE	#Kernel	Norm. MSE
LinChirp	1.46 ± 0.28	7.0	1.00 ± 0.15	21.5	0.92 ± 0.20
Wave	0.98 ± 0.06	5.5	0.73 ± 0.10	20.6	0.79 ± 0.07
Blocks	1.96 ± 0.14	6.0	2.11 ± 0.12	19.4	1.94 ± 0.13
Spike	6.85 ± 0.68	6.1	6.97 ± 0.84	12.8	5.58 ± 0.84

Table: Normalized Mean Square error averaged over 20 runs.

Conclusion on multiple kernel (MKL)

- ▶ MKL: Kernel tuning, variable selection. . .
 - ▶ extention to classification and one class SVM
- ▶ SVM KM: an efficient Matlab toolbox (available at MLOSS)²
- ▶ Multiple Kernels for Image Classification: Software and Experiments on Caltech-101³
- ▶ new trend: Multi kernel and Multi task

²<http://mloss.org/software/view/33/>

³<http://www.robots.ox.ac.uk/~vgg/software/MKL/>

Road map

1 Introduction

2 Tuning the kernel: MKL

- The multiple kernel problem
- SimpleMKL: the multiple kernel solution

3 Non positive kernel

- NON Positive kernels
- Functional estimation in a RKKS
- Non positive SVM

4 Distribution shift

- Distribution shift: the problem
- Density ratio estimation principle

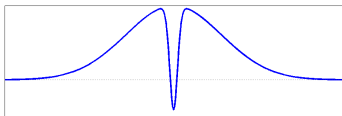
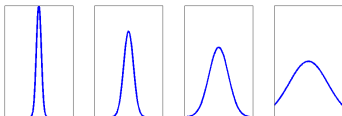
5 Conclusion

Learning with non positive kernel: why?

- ▶ multiple non positive kernels

$$k(\mathbf{x}, \mathbf{x}') = \sum_{m=1}^M d_m k_m(\mathbf{x}, \mathbf{x}')$$

without $d_m \geq 0$,



$$k = m_1 k_1 + m_2 k_2 + m_3 k_3 + m_4 k_4$$

$$m_2 = m_3 = 0 \text{ and } m_1 < 0$$

- ▶ Biological non positive kernels
- ▶ Positive radial kernels are Localized
- ▶ $\tanh(\mathbf{w}^\top \mathbf{x})$

NON Positive kernels

Definition of the associated pre Krein space

- ▶ $\mathcal{K}_0 = \{f \in \mathbb{R}^{\mathcal{X}} | f(x) = \sum_{i=1}^n \alpha_i k(x, x_i), \alpha_i \in \mathbb{R}, x_i \in \mathcal{X}\}$
- ▶ inner product on $\mathcal{K}_0$:
 $f(x) = \sum_{i=1}^n \alpha_i k(x, x_i), \quad g(x) = \sum_{i=1}^m \beta_i k(x, \tilde{x}_i)$

$$\langle f(\cdot), g(\cdot) \rangle_{\mathcal{K}_0} = \sum_{i=1}^n \sum_{j=1}^m \alpha_i \beta_j k(x_i, \tilde{x}_j)$$

the continuity of the evaluation functional

- ▶ $A_x f = f(x) = \langle f(\cdot), k(x, \cdot) \rangle_{\mathcal{K}_0}$ evaluation functional
- ▶ $\langle k(x, \cdot), k(y, \cdot) \rangle_{\mathcal{K}_0} = k(x, y)$ reproducing property

no more norm: $\langle f, f \rangle_{\mathcal{K}_0}$ is NOT always positive

Reproducing Kernel Krein Spaces (RKKS)

Fundamental hypothesis

- ▶ There exist two positive kernels k_+ and k_- such that
$$k(x, y) = k_+(x, y) - k_-(x, y)$$

the RKKS space has to be complete

- ▶ define a topology using the positive kernels k_+ and k_-
- ▶ $\mathcal{K} = \widehat{\mathcal{K}}_0$
- ▶ remark about unicity

Theorem: 3 equivalent statements

- ▶ $\mathcal{K}$ is a RKKS with kernel k
- ▶ $k(x, y) = k_+(x, y) - k_-(x, y)$
- ▶ $k(x, y)$ is dominated by a positive kernel

Examples

- ▶ Minkowski space time

$$\langle (x, y, z, t), (\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t}) \rangle_{\mathcal{K}} = x\tilde{x} + y\tilde{y} + z\tilde{z} - t\tilde{t}$$

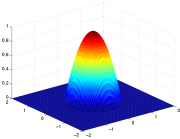
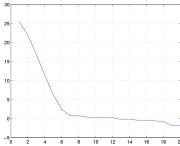
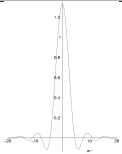
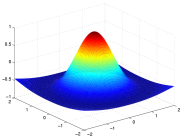
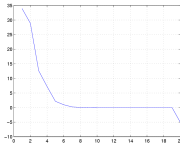
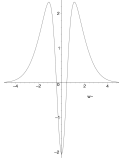
- ▶ Difference of two gaussians

$$k(s, t) := \alpha \exp^{-\frac{\|s-t\|^2}{b}} - \beta \exp^{-\frac{\|s-t\|^2}{c}}$$

- ▶ “Wavelets” kind: assume $\mathcal{H} = V \oplus W$

$$\mathcal{K} = V \ominus W$$

Examples

Kernel	2D kernel	Eigenvalues	Fourier
Epanechnikov kernel $\left(1 - \frac{\ s-t\ ^2}{\sigma}\right)^P,$ for $\frac{\ s-t\ ^2}{\sigma} \leq 1$			
Gaussian Combination $\exp\left(\frac{-\ s-t\ ^2}{\sigma_1}\right) + \exp\left(\frac{-\ s-t\ ^2}{\sigma_2}\right) - \exp\left(\frac{-\ s-t\ ^2}{\sigma_3}\right)$			

Examples of indefinite kernels. Column 2 shows the 2D surface of the kernel with respect to the origin, column 3 shows plots of the 20 eigenvalues with largest magnitude of uniformly spaced data from the interval $[-2, 2]$, column 4 shows plots of the Fourier spectra.

An other view on splines

Interpolation is an ill posed problem

Let $\mathcal{H}$ be a RKHS: Minimize $\|f\|_{\mathcal{H}}^2$ such that $f(\mathbf{x}_i) = y_i, i = 1, n$

In a Krein space

- ▶ NO more norm: how to regularize?
- ▶ project 0 on the set of constrains: $K\alpha = \mathbf{y}$

approximation is an ill posed problem

Let $\mathcal{H}$ be a RKHS: Minimize $\|f\|_{\mathcal{H}}^2 + \frac{C}{2} \sum \xi_i^2$ such that $f(\mathbf{x}_i) - y_i = \xi_i, i = 1, n$ In a Krein space

- ▶ NO more norm: how to regularize?
- ▶ compute a **path** between 0 and the interpolating solution

Non positive SVM: related work

- ▶ considering that the indefinite kernel is a perturbation of a true Mercer kernel.
- ▶ finding a stationary point, which is not unique but each of those performs correct separation. Moreover, it is shown that the problem is then cannot be seen as a margin maximization although a notion of margin can be defined.
- ▶ Krein space instead of a Hilbert space.
- ▶ Applying this to SVM requires to interpret this stabilization setting.
- ▶ a (unconstraint) quadratic program in a Krein space has a unique solution (if the involved matrix is non singular) which is in general a stationary point.

SVM in a Krein space

$$\left\{ \begin{array}{ll} \min_{f, \alpha_0} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{with} & y_i(f(\mathbf{x}_i) + \alpha_0) \geq 1 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{ll} \text{stab}_{f, \alpha_0} & \frac{1}{2} \langle f, f \rangle_{\mathcal{K}} \\ \text{with} & y_i(f(\mathbf{x}_i) + \alpha_0) \geq 1 \end{array} \right.$$

The representer theorem holds: $\langle f, f \rangle_{\mathcal{K}} = \alpha^\top K \alpha$

solve the problem using normal residuals (ie. solving $Ax = b$ via $A^\top Ax = A^\top b$).

$$\begin{aligned} \alpha^\top G &= \mathbf{1} - \lambda \mathbf{y}^\top - \mu^\top + \eta^\top \\ \alpha^\top G G^\top &= (\mathbf{1} - \lambda \mathbf{y}^\top - \mu^\top + \eta^\top) G^\top \end{aligned}$$

This can be seen as least squares. All the other conditions remain identical.

SVM and KKT conditions of optimality

$$\text{KKT conditions for SVM} \quad \begin{cases} \min_{\alpha} & \frac{1}{2} \alpha^{\top} G \alpha - \alpha^{\top} \mathbf{1} \\ \text{subject to} & \alpha^{\top} \mathbf{y} = 0 \\ \text{and} & 0 \leq \alpha_i \leq C \quad \forall i \in [1..n] \end{cases}$$

The stationarity condition is as follows:

$$-\alpha^{\top} G + \mathbf{1} - \lambda \mathbf{y}^{\top} - \mu^{\top} + \eta^{\top} = \mathbf{0}$$

The primal admissibility is given by

$$\begin{aligned} -\alpha^{\top} \mathbf{y} &= 0 \\ \alpha_i &\leq C \quad \forall i \in [1..n] \end{aligned}$$

The dual admissibility is given by

$$\begin{aligned} \alpha_i &\geq 0 \quad \forall i \in [1..n] \\ \mu_i &\geq 0 \quad \forall i \in [1..n] \\ \eta_i &\geq 0 \quad \forall i \in [1..n] \end{aligned}$$

The complementary conditions are

$$\begin{aligned} -\alpha_i \mu_i &= 0 \quad \forall i \in [1..n] \\ (\alpha_i - C) \eta_i &= 0 \quad \forall i \in [1..n] \end{aligned}$$

Point of view 2 : a stabilization problem

Stabilizing $\mathcal{J}$ is equivalent to minimizing $\mathcal{M}$:

$$\mathcal{J} = \frac{1}{2} \alpha^\top G \alpha - \alpha^\top \mathbf{1} \qquad \mathcal{M}(\alpha) = \langle \alpha^\top G - \mathbf{1}^\top, \alpha^\top G - \mathbf{1}^\top \rangle$$

This provides
$$\begin{cases} \min_{\alpha} & \langle \alpha^\top G - \mathbf{1}^\top, \alpha^\top G - \mathbf{1}^\top \rangle \\ \text{with} & \alpha^\top \mathbf{y} = 0 \\ \text{and} & 0 \leq \alpha_i \leq C \quad \forall i \in [1..n] \end{cases}$$

KKT conditions

The stationarity condition is as follows:

$$(\alpha^\top G - \mathbf{1}^\top + \lambda \mathbf{y}^\top - \mu^\top + \eta^\top) G^\top = \mathbf{0}$$

The primal admissibility is given by

$$-\alpha^\top \mathbf{y} = 0$$

$$\alpha_i \leq C \quad \forall i \in [1..n]$$

$$\alpha_i \geq 0 \quad \forall i \in [1..n]$$

The dual admissibility is given by

$$\mu_i \geq 0 \quad \forall i \in [1..n]$$

$$\eta_i \geq 0 \quad \forall i \in [1..n]$$

The complementary conditions are

$$-\alpha_i \mu_i = 0 \quad \forall i \in [1..n]$$

$$(\alpha_i - C) \eta_i = 0 \quad \forall i \in [1..n]$$

Point of view 3 : The projection

Chasing the the most stable point, ie. the admissible point minimizing the gradient of the cost function (which is $\alpha^\top G - \mathbf{1}^\top$).

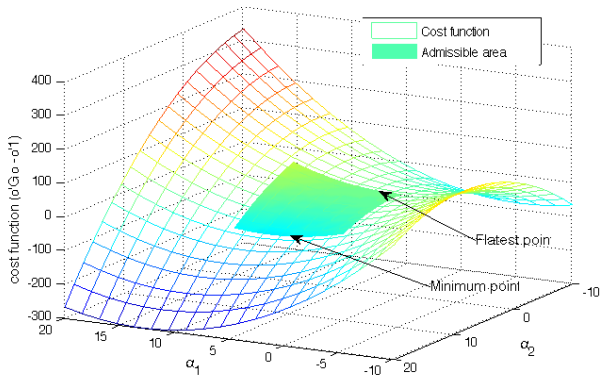


Figure: SVM cost function with sigmoid kernel, illustrated for 2 support vectors. The plain area shows the admissible solutions.

The solver

The proposed algorithm is derived from active set approach for SVM, The sets of points are defined according to the complementarity conditions (see table 2).

Table: Definition of groups for active set depending on the dual variable values

Group	α	η	μ
I_0	0	0	> 0
I_C	C	> 0	0
I_w	$0 < \alpha < C$	0	0

By default, all training points are in the non support vector set I_0 except for a couple with opposite labels which is in I_w . Any other initial situation based on warm-start or a priori does not change the algorithm.

Relaxing constraints in l_0 or l_C

If the current solution is admissible, we check the stationarity conditions for l_0 and l_C . The most violating point is transferred from its group to l_w .

The NPSVM algorithm

- 1: Initialize (one random point for each class in l_w , all others in l_0)
- 2: **while** solution is not optimal **do**
- 3: solve linear system
- 4: **if** primal admissibility is not satisfied **then**
- 5: project solution in the admissible domain : remove a support vector from l_w (to l_0 or l_C)
- 6: **else if** stationarity condition is not satisfied **then**
- 7: add new support vector to l_w (from l_0 or l_C)
- 8: **end if**
- 9: **end while**

Experimental results

- ▶ sigmoid kernel (tanh) : $k(x_i, x_j) = \tanh(scale \times \langle x_i, x_j \rangle + bias)$
- ▶ the epanechnikov kernel: $k(x_i, x_j) = \max(0, 1 - \gamma \langle x_i, x_j \rangle)$

Validation protocol

- ▶ split randomly the dataset, 2/3 for cross validation, 1/3 for test.
- ▶ perform 10 fold cross validation on the validation set
($C \in [0.01, 0.1, 1, 10, 100, 1000]$,
 $\sigma \in [0.10.5, 1, 5, 10, 15, 25, 50, 100, 250, 500] * \sqrt{n}$ for rbf kernel,
 $scale = [pow2(-5 : 1.5 : 2), -pow2(-5 : 1.5 : 2)]$ and
 $bias = [pow2(-5 : 1.5 : 2), -pow2(-5 : 1.5 : 2)]$ for tanh kernel).
- ▶ train the svm on the full validation set with the parameters providing the best average performance during cross validation.
- ▶ test on the separate test set.

Experimental results

Solver	Kernel	Checkers	Checkers 10	Clown
C-SVM	rbf	87.1 % (51 sv)	79.5 % (151 sv)	99.93% (17 sv)
NPSVM	rbf	88.9 % (167 sv)	81.9 % (170 sv)	99.93% (53 sv)
Constraint-NPSVM	rbf	87.1 % (107 sv)	81.6 % (141 sv)	99.97% (54 sv)
NPSVM	tanh	74.9 % (35 sv)	71.8 % (41 sv)	99.97% (81 sv)
Constraint-NPSVM	tanh	86.6 % (114 sv)	81.6 % (145 sv)	99.87% (27 sv)
NPSVM	epanech	86.8 % (133 sv)	81.4 % (118 sv)	99.93% (32 sv)
Constraint-NPSVM	epanech	83.0 % (119 sv)	77.7 % (133 sv)	99.63% (56 sv)

Table: Results on synthetic dataset. Dataset sizes : 200 training points, 3000 testing points. Checkers 10 is a checker dataset with 10% of overlapping between classes. Clowns is also known as apple/banana.

Table: Results on some UCI dataset.

Solver	kernel	Heart	Sonar	Breast
C-SVM	rbf	82.22% (23.2 sv)	84.78% (90.3 sv)	97.47% (53.8 sv)
NPSVM	rbf	83.44% (35.9 sv)	86.09% (94.9 sv)	97.37% (56.6 sv)
NPSVM	tanh	82.44% (14.8 sv)	84.06% (70.1 sv)	97.76 % (116 sv)

Comparison with libSVM

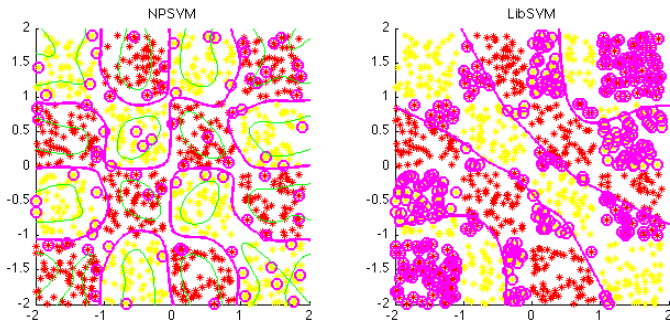


Figure: Results on checkers with NPSVM on the left and libSVM on the right, for an identical sigmoid kernel (scale = 2, bias = -2). Circles are support vectors.

Comparison to IndefiniteSVM

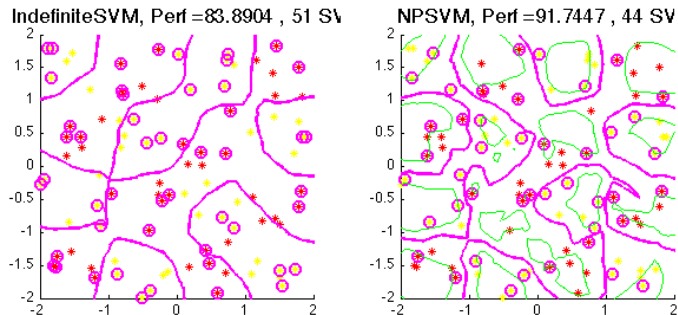


Figure: Results on checkers with IndefiniteSVM on the left and NPSVM on the right, for an identical epanech kernel. Circles are support vectors.

Discussion

SVM with non positive kernels is possible

- ▶ representer theorem
- ▶ sparse solution
- ▶ efficient solver: NPSVM

SVM with non positive kernel is useful

- ▶ to be prove

Road map

1 Introduction

2 Tuning the kernel: MKL

- The multiple kernel problem
- SimpleMKL: the multiple kernel solution

3 Non positive kernel

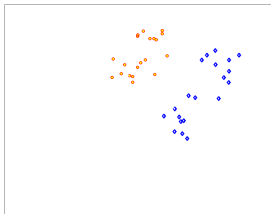
- NON Positive kernels
- Functional estimation in a RKKS
- Non positive SVM

4 Distribution shift

- Distribution shift: the problem
- Density ratio estimation principle

5 Conclusion

Distribution shift: the problem

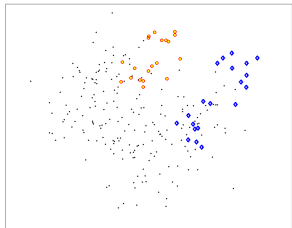


Training data

$$(\mathbf{x}_i^A, y_i^A), i = 1, n$$

i.i.d. from

$$\mathbb{P}_A(\mathbf{x}, y) = \mathbb{P}(y/\mathbf{x})\mathbb{P}_A(\mathbf{x})$$



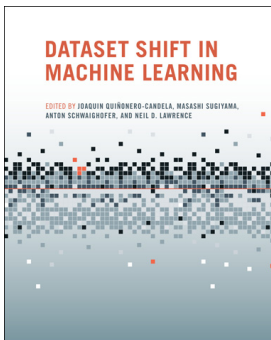
Test data

$$(\mathbf{x}_j^T, y_j^T), j = 1, \ell$$

i.i.d. from

$$\mathbb{P}_T(\mathbf{x}, y) = \mathbb{P}(y/\mathbf{x})\mathbb{P}_T(\mathbf{x})$$

Distribution shift: references



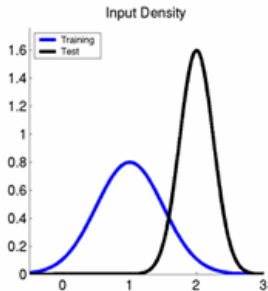
- ▶ Masashi Sugiyama (Tokyo Institute of Technology)
Density ratio estimation methods: Tutorial in
ACML2009^a
- ▶ Arthur Gretton (Max Planck Institute for Biological
Cybernetics) Covariate Shift by Kernel Mean
Matching Workshop at NIPS'09^b
- ▶ Mahesan Niranjan (University of Southampton):
application to intrusion detection

^ahttp://lamda.nju.edu.cn/conf/acml09/files/invited_sugi.pdf

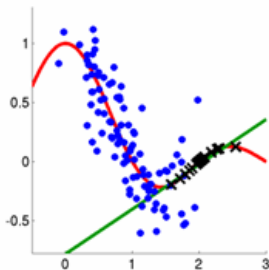
^bhttp://videlectures.net/nipsworkshops09_gretton_cskm/

Covariate Shift

Training and test input follow different distributions, but functional relation remains unchanged.



- Target Function $f(x)$
- Learned Function $\hat{f}(x)$
- Training Sample (x_i, y_i)
- × Test Sample (t_i, u_i)



Goal: Estimate test output from $\{(x_i, y_i)\}_{i=1}^n$

Density ratio estimation principle

$$\begin{aligned}\min_{f \in \mathcal{H}} \mathbb{E}(J(X, Y)) &= \min_{f \in \mathcal{H}} \int_{\mathbf{x}} \int_y J(\mathbf{x}, y) \mathbb{P}_T(\mathbf{x}, y) \, d\mathbf{x} dy \\&= \min_{f \in \mathcal{H}} \int_{\mathbf{x}} \int_y J(\mathbf{x}, y) \mathbb{P}(y|\mathbf{x}) \mathbb{P}_T(\mathbf{x}) \, d\mathbf{x} dy && \text{factorize} \\&= \min_{f \in \mathcal{H}} \int_{\mathbf{x}} \left(\int_y J(\mathbf{x}, y) \mathbb{P}(y|\mathbf{x}) dy \right) \mathbb{P}_T(\mathbf{x}) \, d\mathbf{x} && \text{reorganize} \\&= \min_{f \in \mathcal{H}} \int_{\mathbf{x}} \left(\int_y J(\mathbf{x}, y) \mathbb{P}(y|\mathbf{x}) dy \right) \mathbb{P}_T(\mathbf{x}) \frac{\mathbb{P}_A(\mathbf{x})}{\mathbb{P}_A(\mathbf{x})} \, d\mathbf{x} && \mathbb{P}_A(\mathbf{x}) \neq 0 \\&= \min_{f \in \mathcal{H}} \int_{\mathbf{x}} \left(\int_y J(\mathbf{x}, y) \mathbb{P}(y|\mathbf{x}) dy \right) \mathbb{P}_T(\mathbf{x}) \frac{\mathbb{P}_A(\mathbf{x})}{\mathbb{P}_A(\mathbf{x})} \, d\mathbf{x} \\&= \min_{f \in \mathcal{H}} \int_{\mathbf{x}} \left(\int_y J(\mathbf{x}, y) \mathbb{P}(y|\mathbf{x}) dy \right) w(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) \, d\mathbf{x} && w(\mathbf{x}) = \frac{\mathbb{P}_T(\mathbf{x})}{\mathbb{P}_A(\mathbf{x})}\end{aligned}$$

Importance weighting

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n J(\mathbf{x}_i, y_i) w(\mathbf{x}_i) \qquad w(\mathbf{x}_i) = \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)}$$

Density ratio estimation principle

the algorithm

1. estimate $w(\mathbf{x}_i) = \frac{P_T(\mathbf{x}_i)}{P_A(\mathbf{x}_i)}$
2. solve a weighted version of our favorite learning algorithm

$$(\mathcal{PW}) \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \|f\|^2 + \frac{C}{p} \sum_{i=1}^n w(\mathbf{x}_i) \xi_i^p \\ \text{with} & y_i(f(\mathbf{x}_i) + \alpha_0) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, n \end{array} \right.$$

$p = 1$: L1 weighted SVM

$p = 2$: L2 weighted SVM

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top G \alpha + \alpha^\top \mathbb{I} \\ \text{with} & \alpha^\top \mathbf{y} = 0 \\ \text{and} & 0 \leq \alpha_i \leq C w_i \quad i = 1, n \end{array} \right.$$

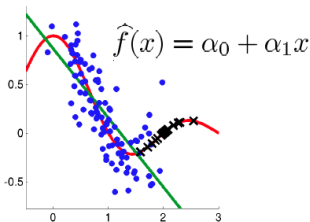
$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top \left(G + \frac{1}{C} W \right) \alpha + \alpha^\top \mathbb{I} \\ \text{with} & \alpha^\top \mathbf{y} = 0 \\ \text{and} & 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

Adaptation Using Density Ratios

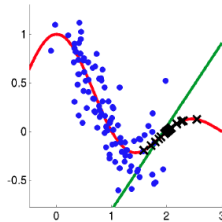
Shimodaira (JSPI2000), Sugiyama & Müller (ICANN2005, Stat&Deci2005)

- Ordinary least-squares is **not consistent**.
- Density-ratio weighted least-squares is **consistent**.

$$\min_{\alpha} \left[\sum_{i=1}^n \left(\hat{f}(x_i) - y_i \right)^2 \right]$$



$$\min_{\alpha} \left[\sum_{i=1}^n \frac{p_{\text{test}}(x_i)}{p_{\text{train}}(x_i)} \left(\hat{f}(x_i) - y_i \right)^2 \right]$$



From M. Sugiyama Tutorial in ACML2009

Estimating the weights

$$w(\mathbf{x}_i) = \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)}$$

- ▶ estimating the distributions –BAD–

1. Parzen (or other) estimate on test data $\hat{\mathbb{P}}_T(\mathbf{x})$
2. Parzen (or other) estimate on training data $\hat{\mathbb{P}}_A(\mathbf{x})$
3. $\hat{w}(\mathbf{x}) = \frac{\hat{\mathbb{P}}_T(\mathbf{x})}{\hat{\mathbb{P}}_A(\mathbf{x})}$

- ▶ Direct estimation: $w(\mathbf{x}) = \frac{\mathbb{P}_T(\mathbf{x})}{\mathbb{P}_A(\mathbf{x})} \iff w(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) = \mathbb{P}_T(\mathbf{x})$
 - ▶ Kullback-Leibler Importance Estimation Procedure

$$\min_{\hat{w}} KL(\mathbb{P}_T(\mathbf{x}) \parallel \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x}))$$

- ▶ Least-Squares Importance Fitting

$$\min_{\hat{w}} \mathbb{E}_A \left(\hat{w}(\mathbf{x}) - \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)} \right)^2$$

Estimating the weights

$$w(\mathbf{x}_i) = \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)}$$

- ▶ estimating the distributions **-BAD-**

1. Parzen (or other) estimate on test data $\hat{\mathbb{P}}_T(\mathbf{x})$
2. Parzen (or other) estimate on training data $\hat{\mathbb{P}}_A(\mathbf{x})$
3. $\hat{w}(\mathbf{x}) = \frac{\hat{\mathbb{P}}_T(\mathbf{x})}{\hat{\mathbb{P}}_A(\mathbf{x})}$

- ▶ Direct estimation: $w(\mathbf{x}) = \frac{\mathbb{P}_T(\mathbf{x})}{\mathbb{P}_A(\mathbf{x})} \iff w(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) = \mathbb{P}_T(\mathbf{x})$
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- ▶ Least-Squares Importance Fitting

$$\min_{\hat{w}} \mathbb{E}_A \left(\hat{w}(\mathbf{x}) - \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)} \right)^2$$

Estimating the weights: empirical distribution

- Empirical distributions

$$\hat{\mathbb{P}}_T(\mathbf{x}) = \frac{1}{\ell} \sum_{j=1}^{\ell} \delta_{\mathbf{x}_j}$$

$$\hat{\mathbb{P}}_A(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{x}_i}$$

- test data (T)

$$\min_{\hat{\mathbf{w}}} \int_{\mathbf{x}} \varphi(\hat{\mathbf{w}}(\mathbf{x})) \mathbb{P}_T(\mathbf{x}) d\mathbf{x} \quad \longrightarrow \quad \min_{\hat{\mathbf{w}}} \frac{1}{\ell} \sum_{j=1}^{\ell} \varphi(\hat{\mathbf{w}}(\mathbf{x}_j))$$

- training data (A)

$$\min_{\hat{\mathbf{w}}} \int_{\mathbf{x}} \varphi(\hat{\mathbf{w}}(\mathbf{x})) \mathbb{P}_A(\mathbf{x}) d\mathbf{x} \quad \longrightarrow \quad \min_{\hat{\mathbf{w}}} \frac{1}{n} \sum_{i=1}^n \varphi(\hat{\mathbf{w}}(\mathbf{x}_i))$$

Kullback-Leibler Importance Estimation

$$\left\{ \begin{array}{l} \min_{\hat{w}} \quad KL(\mathbb{P}_T(\mathbf{x}) \parallel \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x})) \\ \text{with} \quad \int_{\mathbf{x}} \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) d\mathbf{x} = 1 \\ \text{and} \quad 0 \leq \hat{w}(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X} \end{array} \right.$$

$$\begin{aligned} KL(\mathbb{P}_T(\mathbf{x}) \parallel \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x})) &= \int_{\mathbf{x}} \mathbb{P}_T(\mathbf{x}) \log \frac{\mathbb{P}_T(\mathbf{x})}{\hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x})} d\mathbf{x} \\ &= \underbrace{\int_{\mathbf{x}} \mathbb{P}_T(\mathbf{x}) \log \frac{\mathbb{P}_T(\mathbf{x})}{\mathbb{P}_A(\mathbf{x})} d\mathbf{x}}_{\text{constant}} - \int_{\mathbf{x}} \mathbb{P}_T(\mathbf{x}) \log \hat{w}(\mathbf{x}) d\mathbf{x} \end{aligned}$$

$$\left\{ \begin{array}{l} \min_{\hat{w}} \quad - \int_{\mathbf{x}} \mathbb{P}_T(\mathbf{x}) \log \hat{w}(\mathbf{x}) d\mathbf{x} \\ \text{with} \quad \int_{\mathbf{x}} \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) d\mathbf{x} = 1 \\ \text{and} \quad 0 \leq \hat{w}(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X} \end{array} \right. \quad \left\{ \begin{array}{l} \min_{\hat{w}} \quad - \sum_{j=1}^{\ell} \log \hat{w}(\mathbf{x}_j) \\ \text{with} \quad \sum_{i=1}^n \hat{w}(\mathbf{x}_i) = n \\ \text{and} \quad 0 \leq \hat{w}(\mathbf{x}_i) \quad i = 1, n \end{array} \right.$$

Kullback-Leibler Importance Estimation

Use a kernel representation

$$\hat{w}(\mathbf{x}) = \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{x}_i)$$

$$\left\{ \begin{array}{l} \min_{\hat{w} \in \mathcal{H}} - \sum_{j=1}^{\ell} \log \hat{w}(\mathbf{x}_j) \\ \text{with } \sum_{i=1}^n \hat{w}(\mathbf{x}_i) = n \\ \text{and } 0 \leq \hat{w}(\mathbf{x}_i) \quad i = 1, n \end{array} \right. \quad \left\{ \begin{array}{l} \min_{\alpha \in \mathbb{R}^n} - \sum_{j=1}^{\ell} \log K_T \alpha \\ \text{with } \mathbf{e}^T K_A \alpha = n \\ \text{and } 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

Convex (non linear, non quadratic) problem with a sparse solution

Least-Squares Importance Fitting

$$\begin{cases} \min_{\hat{w}} & \mathbb{E}_A \left(\hat{w}(\mathbf{x}) - \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)} \right)^2 \\ \text{with} & \int_{\mathbf{x}} \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) d\mathbf{x} = 1 \\ \text{and} & 0 \leq \hat{w}(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X} \end{cases}$$

$$\begin{aligned} \mathbb{E}_A \left(\hat{w}(\mathbf{x}) - \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)} \right)^2 &= \int_{\mathbf{x}} \left(\hat{w}(\mathbf{x}) - \frac{\mathbb{P}_T(\mathbf{x}_i)}{\mathbb{P}_A(\mathbf{x}_i)} \right)^2 \mathbb{P}_A(\mathbf{x}) d\mathbf{x} \\ &= \int_{\mathbf{x}} \hat{w}(\mathbf{x})^2 \mathbb{P}_A(\mathbf{x}) d\mathbf{x} - 2 \int \hat{w}(\mathbf{x}) \mathbb{P}_T(\mathbf{x}) d\mathbf{x} + \text{constant} \end{aligned}$$

$$\begin{cases} \min_{\hat{w}} & \frac{1}{2} \int_{\mathbf{x}} \hat{w}(\mathbf{x})^2 \mathbb{P}_A(\mathbf{x}) d\mathbf{x} - \int \hat{w}(\mathbf{x}) \mathbb{P}_T(\mathbf{x}) d\mathbf{x} \\ \text{with} & \int_{\mathbf{x}} \hat{w}(\mathbf{x}) \mathbb{P}_A(\mathbf{x}) d\mathbf{x} = 1 \\ \text{and} & 0 \leq \hat{w}(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{X} \end{cases} \quad \begin{cases} \min_{\hat{w}} & \frac{1}{2} \sum_{i=1}^n \hat{w}(\mathbf{x}_i)^2 - \sum_{j=1}^{\ell} \hat{w}(\mathbf{x}_j) \\ \text{with} & \sum_{i=1}^n \hat{w}(\mathbf{x}_i) = n \\ \text{and} & 0 \leq \hat{w}(\mathbf{x}_i) \quad i = 1, n \end{cases}$$

Least-Squares Importance Fitting

Use a feature space representation in a RKHS

$$\hat{w}(\mathbf{x}) = \sum_{k=1}^{\kappa} \alpha_k \phi_k(\mathbf{x})$$

$$\left\{ \begin{array}{l} \min_{\hat{w}} \quad \frac{1}{2} \sum_{i=1}^n \hat{w}(\mathbf{x}_i)^2 - \sum_{j=1}^{\ell} \hat{w}(\mathbf{x}_j) \\ \text{with} \quad \sum_{i=1}^n \hat{w}(\mathbf{x}_i) = n \\ \text{and} \quad 0 \leq \hat{w}(\mathbf{x}_i) \quad i = 1, n \end{array} \right. \quad \left\{ \begin{array}{l} \min_{\hat{w}} \quad \frac{1}{2} \alpha K \alpha - \mathbf{e}^{\top} \Phi_T \alpha \\ \text{with} \quad \mathbf{e}^{\top} \Phi_A \alpha = n \\ \text{and} \quad 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

Convex quadratic program with a sparse solution

Conclusion

- ▶ use weights
- ▶ compute weights at a reasonable cost
- ▶ solve weighted SVM
- ▶ provides a full RKHS embedding with representer theorem

Road map

1 Introduction

2 Tuning the kernel: MKL

- The multiple kernel problem
- SimpleMKL: the multiple kernel solution

3 Non positive kernel

- NON Positive kernels
- Functional estimation in a RKKS
- Non positive SVM

4 Distribution shift

- Distribution shift: the problem
- Density ratio estimation principle

5 Conclusion

what's new since 1995

- ▶ Applications
 - ▶ kernlisation $w^\top \mathbf{x} \rightarrow \langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}} = f(\mathbf{x})$
 - ▶ kernel engineering
 - ▶ sturtured outputs
 - ▶ applications: image, text, signal, bio-info...
- ▶ Optimization
 - ▶ dual: mloss.org
 - ▶ regularization path
 - ▶ approximation
 - ▶ primal
- ▶ Statistic
 - ▶ proofs and bounds
 - ▶ model selection
 - ▶ span bound
 - ▶ multikernel: tuning (k and σ)

challenges: towards tough learning

- ▶ the size effect
 - ▶ ready to use: automatization
 - ▶ adaptative: on line context aware
 - ▶ beyond kenrels: deep learning
- ▶ Automatic and adaptive model selection
 - ▶ variable selection
 - ▶ kernel tuning: coarse-to-fine
 - ▶ hyperparametres: C , duality gap, λ
- ▶ $\mathbb{P}$ change
- ▶ Theory
 - ▶ non positive kernels
 - ▶ a more general representer theorem

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